MIL-C-14887(MU)
24 May 1974
SUPERSEDING
FA-PD-M1-2646
30 April 1965

MILITARY SPECIFICATION

CASE, CARTRIDGE, 106MM, M94B1 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Case, Cartridge 106MM, M94B1 Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.

2.1 (Cont'd)

DRAWING

Armament Command

F8596150	- Case Cartridge 106MM, M94B1
F7548235	- Carton, Ammunition for Interplant Shipment of Steel Cartridge Cases, Empty
IEL-8596150	- Index of Inspection Equipment List

2.2 Other publications.-The following documents form a part of this secification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

ASTM Method E-8	- Tension Testing of Metallic Materials
ASTM Method E-18	- Rockwell Hardness of Metallic Materials
ASTM Method B117	- Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

AWS-A2.0.58 - Welding Symbols, 345 East 47th Street, New York, 17, N.Y.

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Cartridge Case.

3.2.1 Heat treatment and stress relief.-Each cartridge case body (after assembly), shall be austenitized, quenched and tempered or stress relieved uniformly to meet the hardness requirements specified on the applicable drawing.

3.3 Hardness.-Hardness shall be in accordance with the requirements on the applicable drawings and specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.

3.4 Brazed joint.-The head and tube of the cartridge case shall be joined in the manner specified on the applicable drawing. The joint shall withstand a pull of 6500 lbs. minimum, without rupture.

3.5 Strength of weld zone.-The weld zone of the cartridge case body after final assembly and heat treatment shall be as strong as any other parallel section of the body.

3.5.1 The weld zone joint shall be complete and contain no cracks.

3.6 Protective coating.-Protective coating shall be as specified on the applicable drawing and referenced specifications and shall apply to all surfaces of the cartridge case regardless of contour configuration, with the exception of primer hole.

3.7 Maximum profile.-Maximum profile of the cartridge case shall comply with the requirements of the applicable drawings.

3.8 Functional.-The cartridge case shall be subjected to a functional test in the following phases:

Excess Pressure Phase
Cold Temperature Phase
Excess, Excess Pressure Phase (for information only)
Recoil Momentum Phase (for information only)

3.8.1 Soundness.-After firing, the cartridge case shall be free of cracks, splits, bulges or ruptures.

3.8.2 Extraction.-After firing, no cartridge case shall fail to extract, (see 6.3).

3.8.3 Primer retention.-The cartridge case shall retain its primer during firing and extraction.

3.9 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Surface finish.-The requirements for surface finish shall be as specified on the applicable drawing.

3.9.2 Burr.-No cartridge case shall have a burr which might interfere with the assembly of the primer or projectile or which might be injurious to personnel handling the item.

3.9.3 Foreign matter.-No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter prior to inspection and shipping.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.

3.10 Metal defects.-The cartridge case (head & tube) shall be free from die scratches, flutes, folds, dents, cracks, laminations and porosity, inclusions and other metal defects.

3.10.1 Visual standards.-Visual standards for metal defects will be established after the first months production. However, the establishment of visual standards will not waive the intent of the requirement of 3.10. (see 6.4)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification and drawing where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with Specification MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Successful passage of all inspection required by the drawings and specifications shall be considered as a condition precedent to performing Proving Ground functional tests. (See 3.8).

b. Acceptance or rejection of a lot by the Government shall be contingent on successful passage of the Proving Ground function tests.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of two- hundred and thirty (230) cartridge cases as follows:

210 Cartridge cases (coated)
10 Cartridge cases (uncoated)
5 Heads
5 Tubes

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.

4.2.1 (Cont'd)

Two-hundred and five (205) of the above complete and coated cartridge case assemblies shall be forwarded to a designated proving ground for ballistic test. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply. A first article sample or portion thereof as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacture process, material used, drawing, specifications or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions. (see 6.5)

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Heat treatment and stress relief.-Shall contain cartridge cases that have been heat treated or stress relieved in the same equipment, at the same temperature, in one unchanged process. The process thus constituted shall be the basis for selecting the samples for the hardness test hereinafter prescribed.

4.3.2 Examination.-One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis (except as noted) in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination. Four (4X) magnification may be used in performing visual examination. (see 6.6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.

4.3.2.1 Tool control dimensions.-Dimensions marked tool control shall be gaged during First Article testing and after each 250 items produced on each machine in operation and after any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is encountered.

4.3.2.2 Classification of Defects.

4.3.2.2.1 Case, Cartridge, Drawing F8596150 (Prior to Coating).

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR	100% Inspection	
101.	Metal defects	Visual (See 6.4)
MAJOR	AQL .40 Percent	
102.	Diameter of mouth, inside	Gage
103.	Diameter of flange, min.	Gage
104.	Diameter of shoulder, min.	Gage
105.	Diameter of primer hole counterbore	Gage
106.	Thickness of flange, min.	Gage
107.	Thickness of wall at mouth, min.	Gage
108.	Thickness of wall for min. length	Gage
109.	Thickness of wall near head	Gage
110.	Thickness of head	Gage
111.	Depth of primer hole counterbore, min.	Gage
112.	Length from shoulder forward of flange to bottom of primer hole counterbore, min.	Gage
113.	Length from mouth to rear of taper or radius	Gage
114.	Weld not flush with inside diameter of mouth for indicated length	Visual
115.	Radii and chamfers on head missing or incorrect	Visual (See 6.2)
MINOR		
201.	Diameter of undercut under shoulder, min. (if applicable)	Gage
202.	Length total	Gage
203.	Width of undercut, max. (if applicable)	Gage
204.	Thickness through head, at primer hole	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.

4.3.2.2.1 (Cont'd)

Defect	Method of Inspection	
MINOR (Cont'd)		
205. Location of 1st row circumferential holes	Gage	
206. Spacing and alignment of holes, improper	Visual	
207. Hole or holes missing	Visual	
208. Welding extended into perforated area	Visual	
209. Surface finish, improper	Visual	(See 6.2)
210. Marking missing, incorrect or illegible	Visual	
211. Burr	Visual	
212. Foreign matter	Visual	

4.3.2.2.2 Case, Cartridge, Drawing F8596150 (after coating.)

CRITICAL

None defined

MAJOR 100% Inspection

101. Metal defects	Visual	(See 6.4)

MAJOR AQL = .40 Percent

102. True position counterbore with primer hole	Gage	
103. True position primer hole with body	Gage	
104. Diameter of primer hole	Gage	
105. Taper of primer hole incorrect	Gage	
106. Case dented	Visual	

MINOR

201. Protective coating inadequate or damaged	Visual	
202. Varnish - in primer hole	Visual	
203. Compound missing from primer hole	Visual	
204. Foreign matter	Visual	

4.3.2.2.3 Carton, Packed (Prior to sealing), Drawing F7548235.

CRITICAL

None defined

MAJOR

None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.

4.3.2.2.3 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Support missing or damaged	Visual
202. Separator missing or damaged	Visual
203. Improper assembly	Visual
204. Incorrect number in carton	Visual

4.3.2.2.4 Packed Carton (After sealing), Drawing F7548235.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Carton damaged, cut or gouged through all layers	Visual
202. Carton not completely sealed	Visual
203. Contents loose	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection Testing.-The following tests shall be performed: (See 6.5)

4.3.3.1 Maximum profile gage (Major defect).-Each cartridge case shall be gaged in a Government certified maximum profile gage. The cartridge case, after application of protective coating shall freely enter the Government certified maximum profile gage. The gate of the gage shall close without interference with the cartridge case. Failure of the cartridge case to comply with the requirements of 3.7 shall be cause for rejection of the cartridge case.

4.3.3.2 Mouth, inside diameter (major defect).-After maximum profile gaging, the mouth of each cartridge case shall be gaged for the minimum length required by the applicable drawing. The gage shall enter and be withdrawn freely without damage to the varnish. Failure of the cartridge case to comply with the requirements shall be cause for rejection of the case.

4.3.3.3 Mechanical properties.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.

4.3.3.3.1 Hardness.-Ten cartridge cases from the initial production sample and 4 cartridge cases from each production lot shall be subjected to a hardness test, with a Rockwell Hardness Tester or any other method approved by the procuring activity. The test shall be performed in locations as specified on the applicable drawing. Failure of one or more cartridge cases to comply with the requirements of the applicable drawing, shall be cause for rejection of the lot represented by the sample.

4.3.3.3.2 Heat treatment and stress relief (Major defect).-Each cartridge case shall be subjected to a non-destructive hardness comparator test. Failure of the cartridge case to comply with the requirements of 3.2.1 or indications that the cartridge case failed to harden shall be cause for rejection of the cartridge case.

4.3.3.4 Strength of brazed joint (head to tube) (Major defect).-Ten cartridge cases from the initial production sample and 4 cartridges per 8 hr. shift from each brazing machine in operation shall be subjected to this test. The cartridge cases selected shall be mounted on an approved test fixture. The pressure shall be applied gradually until the required load has been attained. Upon removal from the test fixture, the joint will be visually examined for crack(s) in the brazed area or deformation of the cartridge case in the heat affected zone. Failure of one or more cartridge cases to comply with 3.4 shall be cause for rejection of the production on the brazing machine represented by the failure.

4.3.3.5 Soundness of weld (Critical defect).-Each cartridge case, after final expansion but prior to application of the varnish, shall have both surfaces of the weld examined visually to determine compliance with 3.5.1. The weld seam on the outside surface shall be visually examined for crack(s) or other discontinuities. The inside surface of the weld seam shall be examined for crack(s) or other discontinuities, utilizing a probe light (100 ft. Candles min.). In addition a four power borescope shall be used to inspect five cases per hour per welding machine in operation. In the event a defective weld is found, all cases processed since the last acceptable hours examination will be screened using the borescope. If one or more defects are discovered in the back screening of the last acceptable hours production, continued back screening using the borescope will be in effect until an acceptable hours production is reached, i.e., no defects discovered in the screening operation. In addition, the first hours production since the first defect was found shall be subjected to inspection by use of the borescope. If no defects are found during this hour's production, the contractor may return to sample of five (5) per hour. Failure to comply with the requirements shall be cause for rejection of the cartridge case.

4.3.3.6 Protective coating.-The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking action prescribed therein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.

4.3.3.7 Proving ground functional tests.-The following proving ground functional tests will be performed by the Government:

4.3.3.7.1 First Article Sample.

4.3.3.7.1.1 Initial production sample.-The quantities listed in Table I will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table III, except that no re-test will be permitted.

TABLE I

Test Phase		Quantity
Excess Pressure Phase		100
Cold Temperature Phase		50
Excess, Excess Pressure Phase	(Information Only)	50
Recoil Momentum	(Information Only)	5

4.3.3.7.2 Production lot.-The quantities listed in Table II will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table III. The lot will be considered eligible for re-test, using double the number of samples, provided that after firing, the rejection criteria has not been attained.

TABLE II

Test Phase		Quantity
Excess Pressure Phase		40
Recoil Momentum	(Information Only)	5

TABLE III

Defect characteristic	First Sample		Cumulative Sample	
	Accept	Reject	Accept	Reject
1. Splits or Cracks	0	1	Not permitted	
2. Extraction	0	1	Not permitted	
3. Primer retention	0	2	1	2

4.3.3.7.3 Excess pressure phase.-The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be adjusted to obtain 109 to 117 percent of the upper pressure limit of the propellant used with the applicable cartridge case. The complete round will be temperature conditioned at plus 65 degrees Fahrenheit (F) to plus 75 degrees F, and fired in a gun for which the round is standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.

4.3.3.7.4 Cold temperature phase.-The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce a chamber pressure as required by the applicable drawing piezo equivalent pressure, as measured by a copper crushed gage, when temperature conditioned at 70 degrees F. The complete round will be temperature conditioned at minus 40 degrees ± 5 degrees F. and fired in a gun for which the round is standard. The weapon will be operated in the service manner.

4.3.3.7.5 Excess, excess pressure phase.-The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the maximum chamber pressure permitted by the Proving Ground. The complete round will be temperature conditioned at plus 65 degrees F. to plus 75 degrees F. and fired in a gun for which the round is standard. The weapon will be operated in the service manner.

4.3.3.7.6 Recoil momentum phase.-The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the service velocity and pressure for the complete round. The recoil momentum will be recorded.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A - Not applicable.

5.1.2 Level B - Not applicable.

5.1.3 Level C.- The cartridge case shall be packed in accordance with the requirements of Drawing F7548235.

5.2 Labeling and marking.-Labeling and marking for shipment shall be in accordance with MIL-STD-1169.

5.3 Data cards.-Data card information shall be as specified in MIL-STD-1167 Preparation of Ammunition Data Cards.

6. NOTES

6.1 Prior to fabrication of an initial production sample, the contractor upon approval of the contracting officer, may make and submit for evaluation up to 5 process control lots. These process control lots shall consist of not less than 10 nor more than 25 cartridge cases each. The contractor shall furnish all information pertaining to the manufacture of the lots to the contracting officer. All costs for testing these process control lots will be borne by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.

6.2 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.3 Extraction.-The term "extraction" shall mean that the fired cartridge case is loosened and pulled to the rear of the weapon sufficiently to permit removal by hand.

6.4 Visual standards.-Visual standards to be established at contractor's plant in accordance with 3.10.1 or whenever directed by the procuring contracting office.

6.5 Alternative testing procedures.-Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for approval of the responsible technical agency that their effectiveness is equal to or better than the quality provisions specified herein.

6.6 When a product is being inspected in accordance with MIL-STD-1235 and a defective characteristic is found, the product produced prior to the discovered defect will be screened until the frequency (f) number is found to be free of that defect.

6.7 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1315-A250

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:02Z
Check the source to verify that this is the current version before use.